

MEDSENTINEL AI

Multi-Agent AI Battlefield Medical Assistant

A camera or drone scans a casualty scene; MEDSENTINEL automatically triages every wounded person (START protocol), ranks them most-urgent-first, allocates the medical team front-line and back-line, and assembles a virtual specialist consult team around each casualty. A body-worn wearable streams each casualty's live vitals and each medic's physiological strain into the system - so one medic acts with the speed, information and specialist backing of a whole trauma unit.

Project	MEDSENTINEL AI - Project Proposal (v3)
Track	Defence & Security → Military Medical Technology / AI & Autonomous Systems
Output	Working software prototype (live, offline-capable) → edge-deployed field pilot
Live demo	khadejaahmedkz.github.io/medsentinel-ai
Source code	github.com/KhadejaAhmedKZ/medsentinel-ai

Scope & safety. MEDSENTINEL is a decision-support and training/simulation system. It assists a qualified human, who makes every triage and treatment decision. It is medical-only and is **not connected to any weapon, targeting or autonomous-lethality system**. It aligns with the protection of the wounded and of medical data under the Law of Armed Conflict.

1. Executive Summary

Modern battlefield medicine is a race against time and information. Medics operate under extreme stress, often with no medical history, unclear casualty numbers, and time-consuming manual documentation - exactly when correct, rapid prioritisation saves lives. MEDSENTINEL AI is an **agentic system**: a coordinated team of specialised AI agents that acts as a digital extension of the medical team.

The core loop: a camera/drone scans the scene → every casualty is detected and triaged with the international **START** protocol (Immediate / Delayed / Minor / Expectant) → casualties are ranked and the team is allocated front-line vs back-line → each casualty gets an interactive body map that ranks **which wound to treat first** (MARCH order) → a virtual **specialist consult team** assembles based on the injuries. A **wearable** streams live vitals continuously, so triage is hands-free and updates in real time. A working prototype runs today, fully offline, in any browser.

Three design choices make it credible: the triage/clinical logic is an **explicit, auditable rule engine - not a black box**; the system is **offline-first** (essential in comms-denied, electronic-warfare environments); and it is strictly **human-in-the-loop decision support**.

2. Problem Statement

On the battlefield, most deaths are preventable and happen before a medic can reach or process everyone:

- **~87% of combat deaths occur before the casualty reaches a medical facility** - in the medic's hands, not the hospital's (Eastridge et al., *Death on the Battlefield 2001-2011*, J Trauma, 2012).
- Of deaths that were **potentially survivable**, **~90% are from haemorrhage** - treatable with rapid, correctly-prioritised intervention (same study).
- In a **mass-casualty event**, one or two medics face many wounded at once and must decide who to treat first - under fire, with no history.
- **Triage error** (over- and under-triage) is well documented in mass-casualty incidents; wrong prioritisation costs lives and wastes scarce resources.
- **Cognitive overload and decision fatigue** degrade medic performance when stakes are highest.

Statistics are widely-cited figures for the pitch; confirm the exact source before final submission.

Who has this problem: combat medics and Role 1 field teams; medical commanders allocating scarce resources; and - the same pattern - civilian first responders in disasters, bombings and earthquakes (dual-use).

3. Proposed Solution - The Agent Team

MEDSENTINEL deploys a collaborative team of AI agents, each a named specialist. A complex casualty activates several at once, giving the lone medic the coverage of a full unit.

Agent	Role
Scout	Fuses camera/drone imagery to report scene hazards and a safe-approach recommendation, plus a left/centre/right casualty sector breakdown.
Triage	Detects each casualty, applies START scoring (auto-fed by wearable vitals) and ranks them most-urgent-first.
Dr. Sentinel	Per-casualty MARCH clinical prompts; links the wearable's live vitals and the soldier's medical record; flags allergy / drug-interaction risks.
Scribe	Hands-free documentation - voice notes and treatment updates into a structured, exportable casualty report.

Agent	Role
Guardian	Monitors the medic's own physiological strain (wearable HRV/recovery) and prompts recovery to prevent burnout and error.
Overseer	The commander's live picture: casualty counts, team allocation, and a plain-language situation summary.
Logistician	Tracks medical-supply readiness (have vs need) and generates a 9-line MEDEVAC request.
Architect	Compiles an automated after-action review with performance metrics and recommendations.

Specialist Consult Team (assembles per casualty by injury type)

Specialist	Activated by
Vascular & Haemorrhage	Arterial / massive bleeding, amputation
Airway & Anaesthesia	Airway compromise; absent or rapid respiration
Thoracic & Trauma Surgery	Chest / penetrating wounds, pneumothorax
Neurosurgery & Head Injury	Head injury or unresponsiveness
Burns & Plastics	Burns / scald injuries
Orthopaedics	Fractures, deformity, dislocation
General Trauma & Combat Medicine	Always - coordinates the panel (MARCH sweep, hypothermia, evac handover)

4. Wearable Integration - Continuous Vitals (WHOOOP-class)

The soldier and the medic each wear a WHOOOP-class band that **streams** data continuously rather than requiring a tap. This is a stronger design than a passive NFC tag - in one line, **"we replaced a tap with a stream."**

Dimension	NFC chip (previous)	Wearable stream (WHOOOP-class)
How data moves	Medic must physically tap each casualty	Streams continuously - no tap, no contact, hands-free
What you get	A static snapshot (blood type, allergies)	Live, trending vitals: HR, HRV, respiratory rate, SpO ₂ , skin temp, motion
Reachability	Must be beside the casualty	Vitals arrive before you reach them - even behind cover or trapped
Triage	Manual / visual only	Vitals auto-feed START (respiratory rate + perfusion)
Early warning	One-time read	Detects deterioration in real time (falling SpO ₂ , rising HR = occult shock)
Medic wellbeing	Not applicable	Same band on the medic = real HRV/strain for the Guardian agent

It maps onto what we already built. Our triage engine already decides category from respiratory rate, perfusion and responsiveness - exactly what the wearable measures (respiratory rate; HR + SpO₂ as perfusion; accelerometer/motion as an immobility proxy). Integration is a single adapter that maps wearable fields onto the vitals the engine already scores. One device does two jobs: **casualty vitals and medic strain**.

Honest constraint. A consumer wearable's public API is cloud-based and recovery-oriented, and it is not a certified medical device. The tactical version therefore uses **direct BLE ingestion at the edge** (not the cloud), and a **sensor-agnostic** abstraction layer so the same pipeline later accepts military-grade sensors (e.g. Zephyr BioHarness). This de-risks vendor lock-in.

5. System Architecture

Layer	What it does
1. Sensing	WHOOOP-class bands on soldiers (casualty vitals) and medics (strain), over BLE; camera / drone imagery for the scene.
2. Edge gateway	Rugged phone / Nett-Warrior / vehicle node ingests BLE, normalises to a vitals schema, runs the deterministic triage engine LOCALLY , encrypts and buffers - no cloud required.
3. Agent layer	MEDSENTINEL agents consume vitals + imagery: triage, allocation, specialist consult, alerts.
4. Command	Overseer dashboard aggregates live vitals and team status over a secure tactical mesh.
5. Rear sync	When connectivity allows, sync to Role 2/3 for continuity of care and the after-action review.

Data pipeline: WHOOOP BLE → ingestion adapter → vitals schema {deviceId, HR, RR, SpO₂, HRV, skinTemp, motion, timestamp} → map deviceId to soldier profile → triage engine → dashboard + alerting engine → signed audit log.

6. How We Will Implement It

MEDSENTINEL's wearable-vitals layer is **not hypothetical**. A companion prototype - a military biomedical wearable built on the research-grade **EmotiBit** platform with a Python analytics pipeline and a live dashboard - already demonstrates the exact sensing-to-decision chain we need. We **reuse it as our vitals layer** rather than rebuild it, which sharply de-risks the hardware side of the project.

Proven reference implementation (already built)

- **Hardware:** EmotiBit MD board (medical-grade thermopile temperature sensor) on an Adafruit HUZZAH32 (ESP32) Feather, 400 mAh battery, in a custom 3D-printed wrist casing. Streams 16+ body signals - PPG (optical), EDA/GSR, 9-axis IMU and body temperature.
- **Connectivity:** wearable to host over Wi-Fi using OSC over UDP (LSL also supported); Python integration via BrainFlow.
- **Acquisition:** a Python OSC listener maps incoming addresses to variables - heart rate (/HR, validated to 20-250 bpm), temperature (THERM), and a PPG-based SpO₂ estimate.
- **Risk model:** a Random Forest classifier (200 trees, max depth 8, fixed seed; 80/20 stratified split) over four vitals [RR, SpO₂, HR, temperature] outputs a low / medium / high risk class - about 94% accuracy on a human-vital-sign dataset; feature importance ran temperature > respiratory rate > SpO₂ > heart rate; saved as a reusable model file.
- **Dashboard:** a Flask web app with a live streaming endpoint and a risk-interpretation layer (status label, diagnosis text, recommended action). The host does four jobs: network receiver, processing, ML inference, operator display.
- **Architecture:** host-centered and modular - the wearable is the sensing front-end; the laptop / edge node does acquisition, processing, classification and visualisation.

How MEDSENTINEL builds on it

MEDSENTINEL plugs its agent layer directly on top of this pipeline. The vitals it already produces - respiratory rate, SpO₂, heart rate, temperature, plus EDA/GSR and motion - feed straight into two places: (1) our **START triage engine**, which already decides category from respiratory rate, perfusion and responsiveness; and (2) the **Guardian** agent, using EDA/GSR and heart-rate variability for medic stress. The wearable's Random Forest risk class becomes a corroborating **early-warning** signal, while the deterministic START rule remains the auditable authority. The same OSC / BrainFlow ingestion is wrapped in a **sensor-agnostic adapter**, so a WHOOP-class band or a Zephyr BioHarness drops into the same pipeline.

Data pipeline: EmotiBit → Wi-Fi (OSC/UDP) → Python OSC listener (BrainFlow) → normalise {deviceId, HR, RR, SpO₂, HRV, EDA, skinTemp, motion, timestamp} → [Random Forest risk model] + [MEDSENTINEL START engine] → agent layer → dashboard + alerts + signed audit log.

Phased plan

Phase	Goal	Key work
0 - Done	Two proven halves	The MEDSENTINEL multi-agent app (offline, live demo) AND the EmotiBit wearable + Random Forest + Flask dashboard both exist and work.
1	Integrate	Feed the EmotiBit OSC vitals stream into MEDSENTINEL's triage engine; map device to casualty; a live-vitals card replaces the tap.
2	Edge deployment	Move the Python receiver and models onto a rugged offline node; sensor-agnostic adapter (EmotiBit / WHOOP / Zephyr).
3	Hardening	Encrypt the transport (OSC/UDP is plaintext by default), device attestation, EMCON; ruggedise the casing.
4	Field trial	Controlled mass-casualty exercise with EDGE partners; iterate toward certification.

Known constraints (carried into the plan): OSC/UDP is unencrypted by default, so the transport must be wrapped in an encrypted channel (see Cybersecurity); the PPG-based SpO₂ is a prototype-level estimate, not a certified pulse-oximetry reading; and research / consumer wearables are not certified medical devices - hence the Phase-4 certification track.

Stack: EmotiBit + Adafruit HUZZAH32 (ESP32) firmware; Python (BrainFlow, python-osc, scikit-learn Random Forest); Flask streaming dashboard feeding the MEDSENTINEL web UI; rugged edge node. Every agent and sensor is a swappable module.

7. Cybersecurity

A streaming biometric wearable has a larger attack surface than a tap-only tag, and biometric streams are **intelligence-grade data** - they can reveal force size, casualty count, health status and position. We treat every byte as classified.

Threat	Mitigation
Interception (reveals casualties / disposition)	End-to-end AES-256 on the BLE link and transport; strict data minimisation - send only what is needed.
Spoofing / false vitals (mislead the medic)	Sensor fusion - cross-check wearable vs camera triage; physiological sanity-bounds flag impossible values.
Tampering with records (wrong treatment)	Signed, tamper-evident records plus the access audit trail already in the prototype.
RF detection / EMCON (emissions reveal position)	Low-power burst transmission, frequency-hopping, integration with electronic-warfare (SIGN4L); edge processing keeps raw data local.
Device capture (a taken band leaks data)	Remote key revocation / wipe, pre-mission provisioned keys, device attestation.
Unauthorised access	Zero-trust with role-based access (medic / commander) - already implemented in the prototype.

8. Similar Solutions Available Globally

Solution	What it does	How MEDSENTINEL differs
BATDOK (US AFRL)	Medic handheld: patient monitoring, sensor integration, documentation	We add whole-scene camera/drone triage, team allocation and a per-casualty specialist consult panel.
ATAK / TAK medical plugins	Situational-awareness map, medical overlays, MEDEVAC	We add AI triage + wound-priority body maps + agentic clinical support; designed to interoperate with TAK.
Tempus Pro / Philips monitors	Rugged vitals monitoring + telemedicine	Hardware-centric; we are a software multi-agent layer on commodity devices, offline.
Zephyr BioHarness / physiological monitoring	Body-worn vitals used by some militaries	Complementary - it can be our sensor; we own the decision loop and specialist consults.
Academic CV triage / drone detection	Research on detecting/triaging from imagery	We integrate detection into a full operational loop, not a standalone model.

Our differentiator: existing tools monitor one patient or show a map. MEDSENTINEL is the only system that scans the whole scene, prioritises everyone, tells the medic which wound to treat first, and surrounds them with a virtual specialist team - as one integrated, offline-capable, wearable-fed agentic platform.

9. Issues That May Arise (and mitigations)

Issue	Mitigation
CV accuracy in smoke / darkness / occlusion	Multi-sensor fusion (IR/thermal, drone + body-cam), confidence flags, mandatory human confirmation; graceful fallback to manual entry.
Connectivity loss / electronic warfare	Offline-first architecture (already working); sync-when-available.
Model reliability / hallucination	Core triage and clinical logic is deterministic and auditable; the LLM only assists, never decides.
Wearable API / medical-certification limits	Direct BLE edge ingestion instead of the cloud API; sensor-agnostic layer; certification track in Phase 4.
Emissions / EMCON from continuous streaming	Burst + encrypted + frequency-hopping transmission; EW integration.
Data security & privacy of medical data	Encryption, on-device processing, access audit trail, LOAC-aligned handling.
Liability / law of armed conflict	Strictly decision-support, human-in-the-loop; medical-only, never linked to weapons or targeting.
Training-data scarcity	Synthetic data, simulation, exercise/moulage footage, transfer learning.

10. Help Needed (from mentors)

- **Clinical validation** by a military-medical / TCCC subject-matter expert (triage logic, MARCH, specialist recommendations).
- **CV/ML guidance** and access to realistic training/simulation data (synthetic or exercise footage).
- **Wearable / BLE integration** expertise, and a WHOOP (or equivalent) partnership / enterprise access.
- **Edge-hardware & ruggedisation** advice (rugged tablet/wearable, thermal/IR camera, drone feed).
- **Interoperability** help with ATAK/TAK and existing C2 systems, plus secure-comms integration.
- **Field / simulation time** in a realistic mass-casualty exercise, and regulatory / certification guidance.

11. EDGE Group Companies to Support This

Candidate partners (confirm the current EDGE structure before submission):

Company	Capability	Contribution to MEDSENTINEL
ADASI	Autonomous systems / UAVs	Drone/aerial scan feed; autonomous casualty-evacuation tie-in.
KATIM	Ultra-secure communications	Secure transmission of the vitals stream and medical data.
SIGN4L	Electronic warfare / signals	Operating in contested, denied-comms environments; EMCON for continuous emissions.
NIMR	Military vehicles	Integration with casualty-evacuation vehicles / field ambulances.
Bayanat / Space42	Geospatial AI + satellite	Casualty mapping and beyond-line-of-sight connectivity.

Company	Capability	Contribution to MEDSENTINEL
Training & simulation	Exercise capability	A realistic mass-casualty exercise to validate the prototype.

12. Prototype & Links

A working prototype is live today and runs entirely in the browser (offline, no sign-in). It demonstrates all eight agents, the specialist consult team, START triage, the wound-priority body map, the one-by-one 'treat mode' walkthrough, the tourniquet timer, and the commander dashboard.

Live demo	https://khadejaahmedkz.github.io/medsentinel-ai/
Source code	https://github.com/KhadejaAhmedKZ/medsentinel-ai

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